Attention Is All You Need (excerpt for classroom use)

Abstract. We propose a network architecture based solely on attention mechanisms.

3 Model Architecture

3.2 Attention

An attention function maps a query and a set of key-value pairs to an output, where the query, keys, values, and output are all vectors. The output is computed as a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key.

3.2.1 Scaled Dot-Product Attention

The input consists of queries and keys of dimension d_k, and values of dimension d_v. We compute the dot products of the query with all keys, divide each by sqrt(d_k), and apply a softmax function to obtain the weights on the values.

$$\text{Attention}(Q, K, V) = \text{softmax}(Q K^T / \sqrt{d_k}) V \qquad (1)$$

For large values of d_k, the dot products grow large in magnitude, pushing the softmax function into regions where it has extremely small gradients. To counteract this effect, we scale the dot products by 1/sqrt(d_k).

3.2.2 Multi-Head Attention

Instead of performing a single attention function with d_model-dimensional keys, values and queries, we linearly project the queries, keys and values h times.

MultiHead(Q, K, V) = Concat(head_1, ..., head_h) W^O
where head_i = Attention(Q W_i^Q, K W_i^K, V W_i^V) (2)

In this work we employ h = 8 parallel attention layers, or heads. For each of these we use d_k = d_v = d_model / h = 64.

Table 1: Maximum path lengths and per-layer complexity.

Layer Type	Complexity per Layer	Sequential Operations
Self-Attention	O(n^2 * d)	O(1)
Recurrent	O(n * d^2)	O(n)
Convolutional	O(k * n * d^2)	O(1)

Figure 2: (left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several attention layers running in parallel.